

Research-oriented CS student with professional compiler/static-analysis experience and hands-on C++/Python. I design controlled experiments, exact/reference oracles, randomized/metamorphic checks, and reproducible counterexample studies; my strongest current evidence is in algorithms, program analysis, and systems research.

Controlled experiments

UniversalToolchain: frozen corpus, valid controls, post-freeze holdouts, paired comparison, and repeated timing runs.

Mathematical & algorithmic reasoning

PS-form: normalization, modular/GCD reasoning, exact affine analysis, and an independent exact oracle.

C++ / Python research tooling

LLVM, Python harnesses, randomized/metamorphic testing, reproducible counterexamples, and differential execution.

EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer

 - Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#.NET.
- July 1 — August 31, 2026

MCST — Compiler Engineering Intern

 - Implemented an LLVM LICM pass in C++; transformation legality is derived from loop structure, memory accesses, side effects, and speculative-safety constraints.
 - Built a Python harness comparing execution before/after optimization and checking explicit transformation / no-transformation cases; also implemented RPO, Dijkstra, Dinic, and Tarjan SCC in C++23.

PROJECTS

- UniversalToolchain

Current exact test manifest: 1,324/1,324. In a frozen-corpus verifier experiment, B2 detected 32/32 primary and 10/10 challenge operators with 0/100 false positives on valid controls; four post-freeze holdouts were checked separately.
- PS-form Memory Dependence Analyzer

Conservative yes/no/maybe results checked by an independent exact oracle on small domains, randomized/metamorphic tests, and reproducible counterexamples.
- LLVM Interprocedural Global IV Optimization

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.

SKILLS

- Programming

C++23, Python, C17, C#
- Experimental methods

controlled experiments, exact/reference oracles, randomized/metamorphic testing, reproducibility
- Algorithms & CS

data structures, graph algorithms, program analysis, CFG/SSA, compiler optimizations
- Research tooling

Python harnesses, Linux, Git, CMake, counterexample analysis

EDUCATION

HSE University — Software Engineering

RECOGNITION

- Prize-winner in HSE Vysshaya Proba in competitive and industrial programming.

Moscow / remote